



Large Intestine — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A radiology report quotes the typical length of the large intestine. Which value matches the lecture?

EASY

- A. About 10 m
- B. About 0.5 m
- C. About 3.5 m
- D. About 6.5 m
- E. About 1.5 m

2. A histologist examines the mucosa of the large intestine. Which feature would be present?

EASY

- A. Brunner's glands
- B. Crypts of Lieberkühn
- C. Epithelial cells rich in digestive enzymes
- D. Numerous villi
- E. Peyer's patches

3. Which vitamin made by colonic bacteria is needed for blood clotting?

EASY

- A. Thiamine (B1)
- B. Biotin (B7)
- C. Folic acid
- D. Vitamin B12
- E. Vitamin K

4. What proportion of all water absorption in the gut occurs in the small intestine?

EASY

- A. 30%
- B. 50%
- C. 10%
- D. 90%
- E. 70%

5. Between meals, what is the normal state of the ileocecal sphincter?

MEDIUM

- A. Fully relaxed, since cecal distension opens it
- B. Tightly closed, so chyme cannot enter the cecum
- C. Fully open, so chyme enters the cecum rapidly
- D. Open only when the gastrocolic reflex fires
- E. Partially closed, so chyme enters the cecum slowly

6. Immediately after a meal, which event intensifies ileal peristalsis and forces chyme into the cecum?

MEDIUM

- A. The gastroileal reflex
- B. Contraction of the ileocecal sphincter
- C. Distension of the cecum
- D. Haustral churning in the ascending colon
- E. Mass peristalsis in the transverse colon

7. Which pair correctly describes influences on the ileocecal sphincter?

MEDIUM

- A. Gastrin contracts it; cecal distension relaxes it
- B. Gastrin relaxes it; cecal distension intensifies its contraction
- C. Both gastrin and cecal distension relax it
- D. Both gastrin and cecal distension intensify its contraction
- E. Neither influences it; only the gastroileal reflex does

8. What determines the time required for a meal to pass into the colon?

MEDIUM

- A. The number of haustra in the transverse colon
- B. The amount of water absorbed in the ascending colon
- C. Gastric emptying time, because chyme moves through the small intestine at a fairly constant rate
- D. The rate of mass peristalsis in the sigmoid colon
- E. The rate of bacterial fermentation in the cecum

9. A surgeon marks where mass peristalsis normally begins. Which location is correct?

MEDIUM

- A. The rectum
- B. The sigmoid colon
- C. The cecum
- D. About the middle of the transverse colon
- E. The ileocecal valve

10. A patient under extreme parasympathetic stimulation passes ropy mucus with little fecal material as often as every 30 minutes. Stimulation of which nerves increases colonic mucus secretion together with peristalsis?

MEDIUM

- A. The phrenic nerves
- B. The pudendal nerves
- C. The pelvic nerves
- D. The hypoglossal nerves
- E. The superior cervical ganglion

11. A student claims that colonic epithelial cells secrete enzymes to complete carbohydrate digestion. Which correction fits the lecture?

MEDIUM

- A. Enzymes are secreted by villi in the proximal half of the colon
- B. No enzymes are secreted; resident bacteria carry out the final stage of digestion
- C. Enzymes are secreted only when the gastrocolic reflex fires
- D. Enzymes are absorbed from the small intestine and reactivated
- E. Colonic enzymes are secreted, but only in the cecum

12. A patient who ate a large amount of poorly digested carbohydrate has excessive flatus. Which gases result from bacterial fermentation?

MEDIUM

- A. Hydrogen, carbon dioxide and methane
- B. Carbon monoxide and hydrogen sulfide
- C. Only carbon dioxide
- D. Oxygen, nitrogen and ammonia
- E. Indole, skatole and hydrogen sulfide

13. Of the 0.5–1.0 L of water entering the large intestine each day, how much is normally left in the feces?

MEDIUM

- A. About 500 mL
- B. About 300–400 mL
- C. About 100–200 mL
- D. About 0–10 mL
- E. About 1 L

14. A laboratory analyses the composition of feces. Which component is listed in the lecture?

MEDIUM

- A. Sloughed-off epithelial cells from the mucosa
- B. Chylomicrons
- C. Intrinsic factor
- D. Intact villi
- E. Digestive enzymes secreted by the colonic mucosa

15. Which statement correctly contrasts haustral churning with mass peristalsis?

HARD

- A. Churning is a strong wave from mid-transverse colon; mass peristalsis is slow squeezing between adjacent haustra
- B. Churning propels contents into the rectum; mass peristalsis mixes them in the cecum
- C. Both are triggered by the gastroileal reflex and occur continuously between meals
- D. Churning mixes as haustra fill and then squeeze contents onward; mass peristalsis is a strong wave from mid-transverse colon, 3–4 times a day
- E. Churning occurs 3–4 times a day after meals; mass peristalsis is triggered by cecal distension

16. Which sequence best describes the passage of a meal from the stomach to the rectum?

HARD

- A. Gastroileal reflex, mass peristalsis, cecal filling, haustral churning, gastric emptying
- B. Gastric emptying, gastroileal reflex intensifies ileal peristalsis, chyme fills cecum and ascending colon, haustral churning, mass peristalsis into the rectum
- C. Mass peristalsis, haustral churning, gastroileal reflex, cecal filling, gastric emptying
- D. Gastric emptying, haustral churning, mass peristalsis, gastroileal reflex, cecal filling
- E. Cecal filling, gastric emptying, gastroileal reflex, mass peristalsis, haustral churning

17. Some indole and skatole made by colonic bacteria is absorbed rather than passed in feces. What happens to it?

HARD

- A. It is excreted unchanged in the bile as stercobilin
- B. It is converted into vitamin K by hepatocytes
- C. It is stored in the colonic mucosa until the next mass peristalsis
- D. It is carried to the liver, converted to less toxic compounds and excreted in the urine
- E. It is exhaled as hydrogen and methane

18. A patient's colonic bacteria are largely eliminated by prolonged broad-spectrum antibiotics. Which set of functions would be most directly reduced?

HARD

- A. Vitamin K and B vitamin production, fermentation of residual carbohydrate, and conversion of bilirubin to stercobilin
- B. Haustral churning, gastrin release and ileocecal sphincter contraction
- C. Secretion of digestive enzymes, absorption of glucose and formation of haustra
- D. Storage of feces, reabsorption of bile salts and excretion of heavy metals
- E. Mucus secretion, water absorption by osmosis and mass peristalsis

19. After surgery, a patient loses most of the distal half of the colon. Which function is most directly reduced?

HARD

- A. Mass peristalsis beginning in the transverse colon
- B. Secretion of digestive enzymes into the lumen
- C. Storage of feces until a suitable time for excretion
- D. Production of stercobilin in the proximal colon
- E. Most of the large-intestinal absorption of water and ions

20. Which list correctly names substances reabsorbed in the large intestine?

HARD

- A. Water, vitamin A, chylomicrons, urobilinogens, pancreatic enzymes, and toxins
- B. Water, vitamin K, glucose-6-phosphate, indole, intrinsic factor, and mucin
- C. Only water and sodium; vitamins and bile salts are never reabsorbed
- D. Water, bile salts, micelles, gastrin, vitamin D, and stercobilin as the only pigment
- E. Water, vitamin K, biotin (B7) and B5, urobilinogens and stercobilinogens, bile salts, and toxins

✓ Answer key

Q	Answer	Level	Why (lecture slide)
1	E	EASY	The large intestine is about 1.5 m long and 6.5 cm in diameter. [Slide 3]
2	B	EASY	Large-intestinal mucosa has many crypts of Lieberkühn, no villi, and epithelial cells containing almost no enzymes. [Slide 6]
3	E	EASY	Colonic bacteria make some B vitamins and vitamin K, contributing to erythropoiesis and blood clotting. [Slide 4]
4	D	EASY	90% of all water absorption occurs in the small intestine. [Slide 16]
5	E	MEDIUM	Normally the valve remains partially closed so chyme passes into the cecum slowly. [Slide 9]
6	A	MEDIUM	Immediately after a meal, a gastroileal reflex intensifies peristalsis in the ileum and forces chyme into the cecum. [Slide 9]
7	B	MEDIUM	The hormone gastrin relaxes the sphincter; whenever the cecum is distended, its contraction intensifies. [Slide 9]
8	C	MEDIUM	Chyme moves through the small intestine at a fairly constant rate, so the time to reach the colon is set by gastric emptying time. [Slide 10]
9	D	MEDIUM	Mass peristalsis is a strong wave beginning at about the middle of the transverse colon that quickly drives contents into the rectum. [Slide 12]
10	C	MEDIUM	Stimulation of the pelvic nerves markedly increases mucus secretion along with peristaltic motility of the colon. [Slide 7]
11	B	MEDIUM	Mucus is secreted but no enzymes; bacteria ferment remaining carbohydrates and complete digestion. [Slides 6, 13]
12	A	MEDIUM	Bacteria ferment remaining carbohydrates and release hydrogen, carbon dioxide and methane, which contribute to flatus. [Slide 13]
13	C	MEDIUM	All but about 100–200 mL of the water entering the large intestine is normally absorbed by osmosis. [Slide 16]
14	A	MEDIUM	Feces consist of water, inorganic salts, sloughed epithelial cells, bacteria and their products, unabsorbed digested material and indigestible food. [Slide 16]
15	D	HARD	Haustral churning = mixing (haustra distend then contract into the next); mass peristalsis = strong propulsive wave, gastrocolic reflex, 3–4/day. [Slides 11, 12]
16	B	HARD	Gastric emptying sets timing; gastroileal reflex pushes chyme into the cecum; churning mixes; mass peristalsis drives contents into the rectum. [Slides 9–12]
17	D	HARD	The absorbed part is transported to the liver, converted to less toxic compounds and excreted in urine. [Slide 14]
18	A	HARD	Bacteria make vitamin K/B vitamins, ferment carbohydrate and decompose bilirubin to stercobilin (application of the lecture, not stated as a case). [Slides 4, 13, 14]
19	C	HARD	The distal colon functions principally for feces storage (storage colon); most absorption occurs in the proximal half. [Slides 8, 15]
20	E	HARD	Reabsorption includes water, vitamins K, biotin (B7) and B5, urobilinogens/stercobilinogens, bile salts and toxins. [Slide 17]